

MD. TANVIR SHEIKH

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer building LLM applications end to end: LangGraph agents, RAG systems built from scratch and with LangChain, and tool-calling via LLM API's, served with FastAPI and deployed on AWS. Completed an ML internship at FlyRank AI (leakage-audited decline model and capstone). Solid PyTorch and classical ML fundamentals.

EXPERIENCE

Machine Learning Intern – *FlyRank AI*

Jun 2026 – Sep 2026

- **Applied search intelligence to Google Search Ranking & Discoverability**, building a binary classifier (logistic regression, random forest) on ~30K content-page rows across 32 clients to flag pages losing search visibility, using predicted probability as a ranking score for a refresh-review queue.
- **Evaluated with Precision@50** to match reviewer capacity, using client-grouped splits (GroupShuffleSplit) and a data contract that excluded trend-derived and ID features to prevent leakage.
- **Designed a transparent baseline score** with reason codes and audited signal claims against sample-size floors; queried a DuckDB warehouse with NULL-safe SQL.
- **Learned and applied a full workflow** (data contract, signal audit, baseline, validation audit, action playbook) and documented findings and limitations in a capstone paper.

SKILLS

LLM & AI Engineering: LangGraph, LangChain, RAG, Agentic AI, Tool Calling, Prompt Engineering, LLM Evaluation, LLM Fine Tuning, Groq, HuggingFace, ChromaDB, Embeddings, MMR

ML & Deep Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, CNNs, Transfer Learning

Backend & Data: Python, FastAPI, Pydantic, SQL, PostgreSQL, MySQL, Pandas, NumPy

Cloud & Tools: AWS, Docker, Git, Streamlit, Google Colab

PROJECTS

Personal AI Assistant with Corrective RAG – LLM Agent

[GitHub](#) | [Live Demo](#)

- **Built a LangGraph agentic** system that conditionally routes requests across calculator, web-search, and RAG tools, with LLM-based relevance grading and answer verification.
- **Engineered the RAG pipeline:** multi-format ingestion (PDF/DOCX/TXT/MD), query expansion, MMR retrieval, deduplication, and ChromaDB with HuggingFace embeddings; UUID-based isolation keeps each session's knowledge base separate.
- **Implemented production-oriented** reliability features including multi-model Groq fallback routing, Pydantic-validated outputs, FastAPI SSE streaming with cancellation, and deployed on AWS EC2 using Docker.

Tech Stack: Python, LangGraph, LangChain, FastAPI, ChromaDB, HuggingFace, Groq, Pydantic, AWS, SSE

Custom RAG Pipeline — Built from Lower-Level Components

[GitHub](#)

- **Built RAG from first principles** with custom embedding, vector-store, and retriever classes instead of LangChain abstractions.
- **Implemented ingestion to generation** (PyMuPDF, recursive chunking, ChromaDB, Llama 3.3 70B via Groq); debugged a ChromaDB distance-metric inversion and a silent fallback to default embeddings.

F1 Pit Stop Prediction — Kaggle Competition

[GitHub](#) | [Kaggle](#)

- **Built a stacked ensemble** (LightGBM, XGBoost, CatBoost, PyTorch ANN feeding a Ridge meta-learner) with GroupKFold validation and domain-specific feature engineering, scoring 0.95167 on the private leaderboard.

Attendance by Selfie — CNN-Based Attendance Platform

[GitHub](#) | [Live Demo](#)

- **Built an end-to-end system** for attendance via classroom photo or voice, using dlib face recognition, Resemblyzer speaker verification, and a Supabase (PostgreSQL) backend with role-based portals; fixed schema and data-integrity issues.

Deep Learning Projects — PyTorch & NumPy

[CIFAR-10](#) | [Emotion Detector](#) | [MNIST](#) | [Crack Detection](#)

- **CIFAR-10 CNN:** ~87.75% validation accuracy, 0.90050 on Kaggle.
- **Facial Emotion Detector:** custom CNN vs fine-tuned ResNet-18 on FER-2013 (7 classes).
- **MNIST from scratch:** pure-NumPy network with hand-derived backpropagation.
- **Surface Crack Detection:** PyTorch CNN based with ~99.8% validation accuracy on Kaggle

CERTIFICATIONS

Course name: Complete AI/ML Engineering Course by Apna College (Prime Batch 1.0)

Nov 2025–Apr 2026

Technologies learned: Python, Data Pre-processing, Data Visualization, Math for AI (Statistics & Probability), Machine Learning (Supervised: Regression, Classification; Unsupervised: Clustering, Association; Reinforcement Learning), Logistic Regression, SVM, Decision Tree, K-Means Clustering, Precision/Recall/F1, Bias-Variance Tradeoff, Deep Learning, Perceptron, FNN/CNN/RNN Architectures, Transformers, PyTorch, GenAI, LLMs, NLP, RAG, GAN, OpenAI APIs, FastAPI (AI development), SQL for Data Science, Git & GitHub, Docker

EDUCATION

BSc in Computer Science and Engineering — Alhaj Mockbul Hossain University College **Apr 2022 – Expected Nov 2026**

Relevant Coursework: Database Management Systems, Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Software Development Engineering.